WEEKLY TELECOM, 5G & RAN INTELLIGENCE

HIKMAH *Signal*

Architecture & Operations — how this autonomous newsletter is built and run

Domain: Network Intelligence for Telecom Engineers | Auto: Tuesday 06:00 GST | Vol. II · GCC & Global Edition

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01 What this does

HIKMAH Signal is a fully autonomous weekly intelligence newsletter. Every week a CrewAI agent crew reads 15 curated industry sources, removes anything already covered, scores what remains for a **senior audience** (engineers, architects, consultants, executives), and publishes the strongest **24 stories** — 6 in each of 4 sections — as desktop and mobile web pages and PDFs, archived on GitHub Pages.

It is not written for juniors: each story carries a decision-grade **"Signal"** that fuses the architectural implication with the strategic/business consequence — adoption timing, cost, risk, and the move a senior leader should make.

02 How CrewAI works here

CrewAI orchestrates a small team of role-specialised LLM agents that hand work to each other in a fixed sequence (a "crew"). Each agent has a role, a goal, a model, and optionally tools it can call. Output of one task becomes the context of the next.

RSS Sources 15 feeds → **Scout** fetch · dedup · classify → **Analyst** score · write Signal → **Publisher** compile JSON → **Render** HTML + PDF → **Publish** GitHub + email

03 The agents

#	AGENT	MODEL	TOOLS	RESPONSIBILITY
01	**Data Scout**	claude-haiku-4-5	fetch_rss, dedup	Fetches every RSS feed, removes already-seen articles via SHA-256 dedup (persisted in a per-project SQLite DB), and classifies the rest into the 4 sections.
02	**Principal Analyst & Strategist**	claude-sonnet-4-6	(reasoning only)	Scores each article 0-100 for senior-audience significance, drops anything below 65, and writes the 3-sentence summary plus the decision-grade 'Signal'. Keeps the top 6 per section.
03	**Intelligence Publisher**	claude-sonnet-4-6	(reasoning only)	Compiles the final, schema-valid JSON payload (issue metadata, stats, ticker, 4 sections x 6 entries) that drives the templates. Emits JSON only.

Models are addressed as anthropic/claude-* so CrewAI routes them to Anthropic. The Scout uses the fast Haiku model for high-volume triage; the Analyst and Publisher use Sonnet for judgement and strict formatting.

04 Context & data flow

- **Tools as context:** the Scout's fetch_rss tool returns raw articles; its dedup tool checks each URL's SHA-256 against a per-project SQLite DB so a story is never repeated across issues.
- **Task chaining:** each agent's output is passed as the *context* of the next task — Scout→Analyst→Publisher — so the Analyst sees the classified set and the Publisher sees the scored set.
- **Structured payload:** the Publisher emits one JSON object — issue metadata, stats, ticker headlines, and 4 sections x 6 entries (title, url, source, date, summary, signal, keywords, vendors, score).
- **Branding merged in code:** the project's colours, wordmark and section taxonomy (from brand.py) are merged into the payload before a shared Jinja2 template renders it.

05 Coverage — the 4 sections

01 · Market & Operators
Coverage · M&A · Spectrum Deals · Vendor Financials
Watching: Ericsson · Nokia · Huawei · ZTE · Regional Operators

02 · Network Infrastructure
5G-Adv · 6G · O-RAN · CU/DU · CloudRAN · Private 5G
Watching: RAN Architecture · Transport · Core · Edge

03 · AI, Standards & Analytics
3GPP Rel-18/19 · GSMA APIs · AI/ML RAN · Ookla
Watching: Machine Learning · Automation · Performance Intelligence

04 · Spectrum, Satellite & Regulation
NTN · ITU · Cybersec · Fiber · Satellite · Regulation
Watching: Policy · Licensing · NTN Integration · Critical Infrastructure

06 The suite it belongs to

This is one of **five independent projects** in the HIKMAH suite, each with its own crew, sources, dedup DB, brand and weekly slot, all sharing one library (hikmah-shared: renderer, PDF generator, GitHub publisher, email sender, dedup):

PROJECT	DOMAIN	DAY (06:00 GST)
signal	Telecom / 5G / RAN	Tuesday
intelligence	AI / Agentic / LLM	Wednesday
dataml	MLOps / Data Science	Thursday
cloudinfra	Cloud / Containers / Edge	Friday
dataarch	Databases / Big Data / GPU / API	Saturday

07 What an "issue" is

Each run produces one numbered issue with **4 deliverables** — published to CrewAI/signal/issues/ and served on GitHub Pages:

- issue-NNN.html — desktop edition (multi-column)
- issue-NNN-mobile.html — phone edition (single column)

- issue-NNN.pdf — A4 print/archive PDF
- issue-NNN-mobile.pdf — phone-format PDF

An issue number auto-increments each full run. Runs can be **manual** (run_now.py signal) or **automatic** (the weekly scheduler), with an offline --demo mode for previewing without an API key.

08 Sources (15)

- lightreading.com
- fiercewireless.com
- fierce-network.com
- fiercetelecom.com
- rcrwireless.com
- capacitymedia.com
- totaltele.com
- datacenterdynamics.com
- ericsson.com
- theregister.com
- nextgov.com
- telecomlead.com
- gsma.com
- opensignal.com
- networkworld.com